

Diamond Surface Strain Pipeline

From Surface Termination to NV Spin Perturbation

1 Motivation

Question. Does the chemical termination of a diamond surface (H, F, ether O, ketone O, bare reconstruction) meaningfully perturb the spin resonance of a near-surface NV centre through strain?

Logic chain.

1. Surface termination imposes a preferred in-plane bond spacing.
2. The bulk diamond lattice is mechanically rigid — the mismatch generates a residual surface stress σ_{ij} .
3. The stress field decays into the crystal, reaching a nearby NV.
4. Strain at the NV shifts the zero-field splitting parameters D and E (§10), through a six-parameter spin-strain Hamiltonian (§12).

Scope. Three surface orientations: (100), (110), (111). A library of 24 validated surface motifs covers the relevant terminations.

2 Computational Setup

Functional. GGA-PBE throughout.

Pseudopotentials (SSSP-style set, verified against run output).

Species	File	Type
C	C.pbe-n-kjpaw_psl.1.1.0.0.UPF	PAW, $Z_{val} = 4$
H	H.ONCV_PBE-1.0.oncvpsp.upf	ONCV norm-conserving, $Z_{val} = 1$

Plane-wave cutoffs. $E_{cut}^{wfc} = 60$ Ry; $E_{cut}^{\rho} = 480$ Ry (8 $\times$ ratio, required with a PAW carbon pseudopotential). Marzari–Vanderbilt (cold) smearing, $\sigma = 0.01$ Ry.

k -point meshes. Each surface orientation exposes a different 2D periodic lattice; density is matched via $n_i \propto 1/|A_i|$ (a longer real-space repeat needs fewer k -points).

Surface	In-plane cell (Å)	k -mesh	n_x/n_y (actual / ideal)
C(100) 2×1	5.054×5.054	$5 \times 5 \times 1$	1.00 / 1.00
C(110) 1×1	2.527×3.574	$9 \times 7 \times 1$	1.29 / 1.41
C(111) 1×1	2.527×2.527 (hex.)	$9 \times 9 \times 1$	1.00 / 1.00

$k_z = 1$ in all cases (no periodicity in z). C(100)’s reconstructed cell is square, so an isotropic mesh already matches the ideal density. C(110)’s rectangular cell is sampled marginally below the ideal ratio (1.29 vs. 1.41) — negligible for stress *differences*, but worth revisiting if absolute stress values are needed.

Python Process Architecture

Every box below is a script in the repository; every arrow label is the file its upstream box writes and its downstream box reads. The bulk-reference branch (left) runs independently and feeds only a_0 forward, into slab construction.

3 Slab Model

A surface is modelled as a periodic slab: periodic in-plane ($\mathbf{A}_1, \mathbf{A}_2$), finite in z , padded with ~ 10 Å of vacuum (9.8–10.2 Å measured across the three faces). The current calculations use $N = 6$ C layers; the bottom 2 are held fixed (`if_pos 0 0 0`) to represent the rigid substrate.

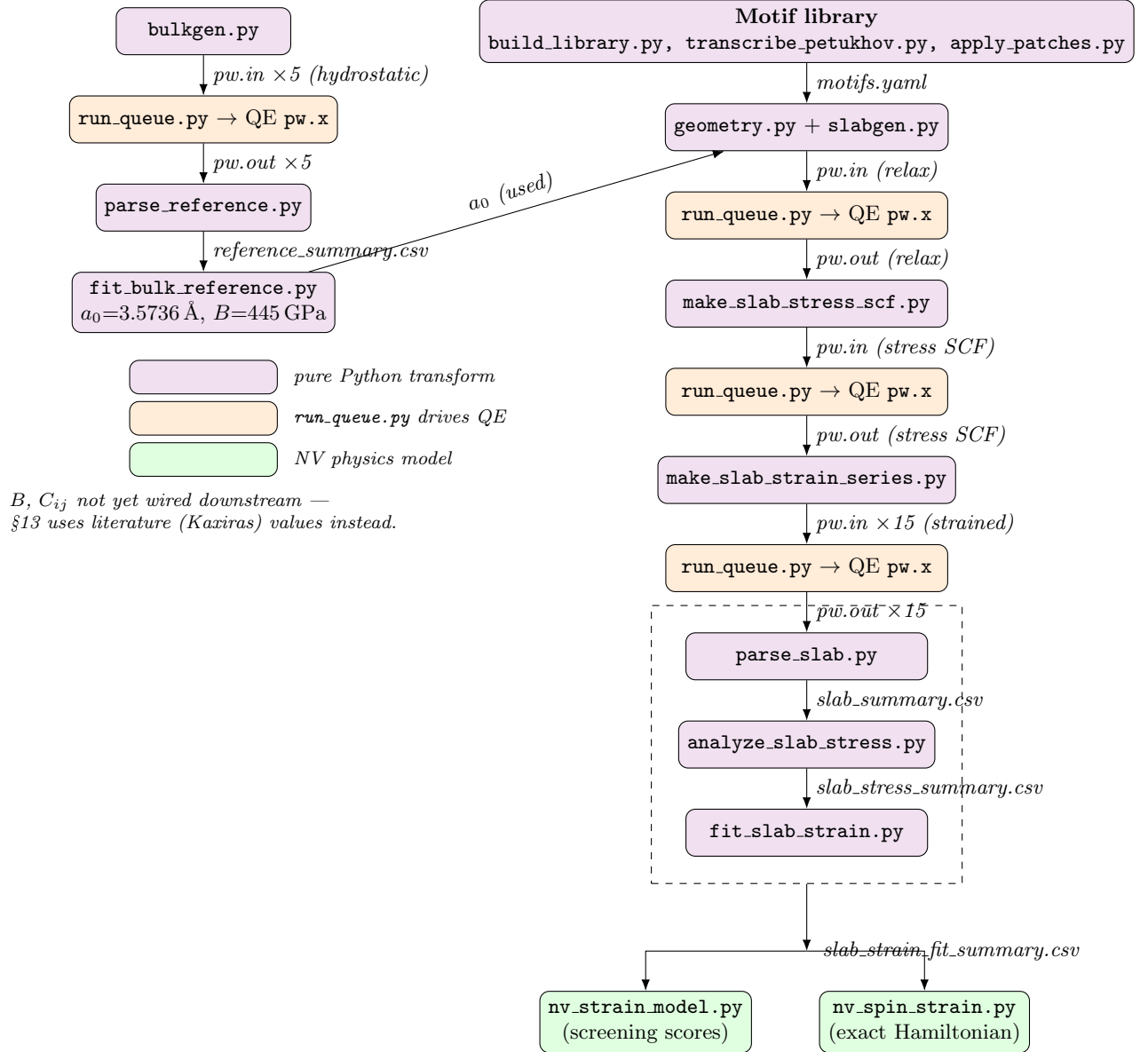


Figure 1: Full data-flow architecture. Left: independent bulk-reference branch (§6), contributing only a_0 to slab construction. Right: the main slab pipeline, from motif library through relaxation, stress extraction, and strain fitting, terminating in the two NV models (§11, §12–§14).

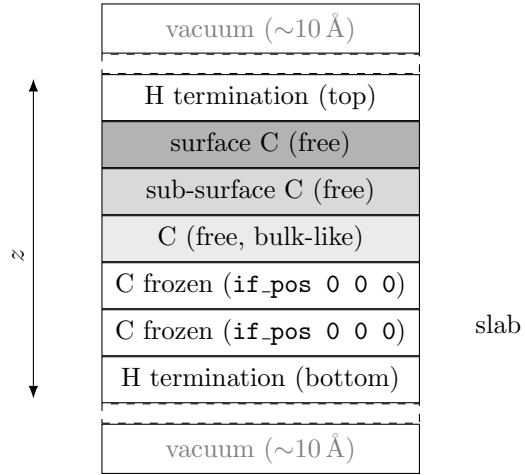


Figure 2: Slab geometry (symmetric monohydride termination). Dark grey: free C layers that relax during DFT optimisation. Light grey: increasingly bulk-like free layers. H termination is applied symmetrically to both faces so the two surfaces are chemically identical.

4 Slab Construction

Algorithm (`geometry.py`, `slabgen.py`). No hand-derived coordinates are used; every slab is generated numerically.

1. Tile the 8-atom diamond cubic cell; rotate so the surface normal is $+z$.
2. Cluster atoms by z into discrete layers. On (111), a parity search selects the single-dangling-bond (SDB) face: each surface C has exactly one missing tetrahedral bond.
3. Apply the surface termination: place each terminating atom along the computed dangling-bond direction at the target bond length.
4. Validate: reject any structure with pentavalent C or a bond $< 1.30 \text{ \AA}$.

Target bond lengths (ideal construction geometry).

Bond	Length ($\text{\AA}$)	Notes
C–C bulk	1.545	$\sqrt{3}/4 \times a_0$
C–H	1.09	monohydride
C–F	1.36	monofluoride
C–O (ether bridge)	1.43	C–O–C
C=O (ketone)	1.21	sp^2 surface C
C–C dimer, bare (100)	1.37	π -bonded
C–C dimer, H-sat (100)	1.60	σ -only

Motif library (`motifs.yaml`). 24 validated motifs are stored in dimensionless units (in-plane fractional coordinates and z/a_0), so the library is valid at any a_0 . Literature reconstructions (Pandey π -bonded chain, Petukhov dimerised graphitic chain) are stored as transcribed coordinates and spliced onto a bulk base slab at generation time.

5 Structure Relaxation

Why. The ideal-constructed slab places surface atoms at bulk-truncated positions — not equilibrium positions; surface forces reach $\sim 100 \text{ meV/\AA}$. The stress tensor of an unrelaxed slab is dominated by construction artefacts and carries no physical meaning.

How (Quantum ESPRESSO `pw.x`, `calculation = 'relax'`).

- **Algorithm:** BFGS quasi-Newton ionic minimiser.
- **Cell:** fixed. In-plane lattice vectors held at the PBE bulk value $a_0 = 3.5736 \text{ \AA}$ throughout — models a thin film on a rigid substrate.
- **Frozen layers:** bottom 2 C layers held fixed.
- **Convergence:**

$$|F_i| < 10^{-4} \text{ Ry/bohr} \quad \forall i \text{ (free atoms)} \quad (1)$$

$$|\Delta E_{tot}| < 10^{-5} \text{ Ry} \quad (\text{successive BFGS steps}) \quad (2)$$

Outcome. Bond lengths, dimer tilting, layer rumpling, and reconstructions all relax; the stress tensor extracted thereafter reflects actual surface physics.

6 Bulk Reference Calculation

Why. PBE overestimates the lattice constant ($a_0^{exp} = 3.567 \text{ \AA}$; $a_0^{PBE} = 3.5736 \text{ \AA}$). All slabs are built at a_0^{PBE} to avoid a spurious experiment-vs-theory strain contaminating every downstream number. The bulk modulus B is also needed to convert strain-fit slopes into physical stress units.

Procedure. A 5-point hydrostatic strain series, $a = a_0(1 + \varepsilon)$, $\varepsilon \in \{-1\%, -0.5\%, 0, +0.5\%, +1\%\}$, SCF at each volume.

Fits. Rather than trust one fit, three independent routes to the same equilibrium point are computed and required to agree — agreement *is* the validation; disagreement would flag a non-parabolic or under-sampled window.

$$E(\varepsilon) = A\varepsilon^2 + B\varepsilon + C \Rightarrow \varepsilon_0 = -\frac{B}{2A} \quad (3)$$

$$P(\varepsilon) = m\varepsilon + b \Rightarrow \varepsilon_0 = -\frac{b}{m}, \quad B = -V_0 \frac{dP}{dV} \quad (4)$$

$\varepsilon_0(E)$ is where the energy parabola is flattest (zero restoring force); $\varepsilon_0(P)$ is where the fitted pressure literally crosses zero. A third route, $E(V)$, gives a_0 independent of the strain convention entirely.

Consistency check: $|\varepsilon_0(E) - \varepsilon_0(P)| \leq 0.003$. **Results:** $a_0^{PBE} = 3.5736 \text{ \AA}$ (all three routes agree to $< 0.001 \text{ \AA}$); $B = 445 \text{ GPa}$ (consistent with the accepted PBE diamond value, $\sim 430\text{--}450 \text{ GPa}$).

7 Surface Stress Extraction

Setup. After relaxation, a single-point SCF is run with `tstress=.true.`. QE reports σ_{ij} in kbar, averaged over the full cell volume (slab + vacuum).

Conversion to 2D surface stress.

$$\tau_{ij} [\text{N/m}] = \sigma_{ij} [\text{kbar}] \times L_z [\text{\AA}] \times 0.005 \quad (5)$$

where $0.005 = (0.1 \text{ GPa/kbar}) \times (0.1 \text{ N m}^{-1}/\text{GPa \AA}) \div 2$ (two faces share the load).

In-plane decomposition.

$$\sigma_{mean} = \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) \quad \text{hydrostatic-like; shifts } D \quad (6)$$

$$\sigma_{anis} = \sigma_{xx} - \sigma_{yy} \quad \text{anisotropy; activates } E \quad (7)$$

Sign. $\tau > 0$ (tensile): surface prefers larger in-plane spacing. $\tau < 0$ (compressive): surface prefers smaller spacing. On (111), C_{3v} symmetry forces $\sigma_{anis} = 0$ identically.

8 In-Plane Strain Series

Why. A single stress measurement at $\varepsilon = 0$ gives b (residual stress) but not the surface stiffness or the zero-stress natural mismatch. A strain series gives both, as a genuine finite-difference measurement rather than an assumption.

Strain application. Starting from the relaxed (stress-SCF) geometry, lattice vectors are scaled rigidly — no ionic re-relaxation after straining:

$$\text{Biaxial: } \mathbf{A}_1 \rightarrow (1 + \varepsilon)\mathbf{A}_1, \quad \mathbf{A}_2 \rightarrow (1 + \varepsilon)\mathbf{A}_2 \quad (8)$$

$$\text{Uniaxial-}x : \mathbf{A}_1 \rightarrow (1 + \varepsilon)\mathbf{A}_1, \quad \mathbf{A}_2 \rightarrow \mathbf{A}_2 \quad (9)$$

Atomic x, y coordinates transform with the new in-plane basis (fractional coordinates preserved); z is unchanged. $\varepsilon \in \{-1.0\%, -0.5\%, 0, +0.5\%, +1.0\%\}$. Three modes are run: biaxial (primary, feeds §12–§14); x - and y -uniaxial (diagnostic decomposition of the anisotropic response, not used as a standalone physical state).

9 Stress–Strain Fitting

Model. Within the linear (Hookean) regime ($|\varepsilon| \leq 1\%$, confirmed by the bulk reference fits in §6):

$$\sigma_{mean}(\varepsilon) = m\varepsilon + b \quad (10)$$

Parameter	Units	Physical meaning
m	kbar	Surface stiffness $d\sigma_{mean}/d\varepsilon$, measured directly as a finite difference for <i>this</i> surface — not assumed from bulk elastic constants, since a reconstructed surface can be stiffer or softer.
b	kbar	Residual stress at $\varepsilon = 0$. Primary input to NV screening (§11) and to the elasticity solve in §13.
$\varepsilon_0 = -b/m$	—	Strain the surface would need to relax to in order to zero its own stress; $\varepsilon_0 > 0$ means the surface prefers larger in-plane spacing than bulk.
τ_{mean}	N/m	b converted via Eq. (5); comparable to experimental surface-stress literature.

Fit quality: $R^2 > 0.98$. A failing R^2 usually means the sampled window left the linear-elastic regime (e.g. a reconstruction distorting under tension), not numerical noise — read it as “check the geometry”, not “add more points”.

10 NV Centre Physics

System. The NV centre is a spin- $S = 1$ point defect ($N_s V^-$). Its three magnetic sublevels ($m_s = 0, \pm 1$) split at zero field.

Leading-order spin Hamiltonian.

$$\mathcal{H} = D \left(S_z^2 - \frac{S(S+1)}{3} \right) + E(S_x^2 - S_y^2) + \gamma \mathbf{B} \cdot \mathbf{S} \quad (11)$$

- $D_0 = 2.870$ GHz — axial zero-field splitting (bulk, C_{3v}).
- $E = 0$ in bulk (forced to zero by C_{3v}).

- $\gamma = 28.0 \text{ GHz/T}$.

Strain perturbations, leading order.

$$\Delta D \propto \varepsilon_{mean} = \frac{1}{2}(\varepsilon_{xx} + \varepsilon_{yy}) \quad \text{one ODMR peak moves} \quad (12)$$

$$E \propto \varepsilon_{xx} - \varepsilon_{yy} \quad \text{peak splits in two} \quad (13)$$

This two-parameter picture is qualitatively correct — it tells you which strain channel drives which observable — but it is only the leading-order limit of a complete six-parameter microscopic Hamiltonian (§12), and it says nothing about how a given crystal strain looks to each of the four inequivalent $\langle 111 \rangle$ NV orientations (§13).

11 NV Screening Model

Risk scores (normalised to dataset maximum, $\in [0, 1]$):

$$\text{D-score} = \frac{|b_{biaxial}|}{\max_{dataset} |b|} \quad \text{risk of } \Delta D \text{ shift} \quad (14)$$

$$\text{E-score} = \frac{|\sigma_{anis}|_{\varepsilon=0}}{\max_{dataset} |\sigma_{anis}|} \quad \text{risk of } E\text{-splitting} \quad (15)$$

$$\text{Overall} = \frac{1}{2}\text{D-score} + \frac{1}{2}\text{E-score} \quad \text{equal weighting} \quad (16)$$

These are deliberately crude, transparent proxies — useful for ranking surfaces *before* paying for the full Hamiltonian treatment of §12, not as calibrated MHz predictions.

Results for H-terminated slabs (corrected, symmetric H/H slabs).

Surface	σ_{mean} (kbar)	σ_{anis} (kbar)	D-score	E-score	Risk
C(100)–H	−21.4	56.8	0.56	1.00	High
C(110)–H	+38.1	−28.2	1.00	0.50	High
C(111)–H	+4.5	0.0	0.12	0.00	Low

These supersede the earlier asymmetric H/bare-slab numbers (now archived, [results/archive_asymmetric_6L/](#)); the ranking is qualitatively unchanged but magnitudes and signs are not directly comparable to that archive.

Key observations.

- C(111)–H: C_{3v} symmetry enforces $\sigma_{anis} = 0$ regardless of σ_{mean} .
- C(100)–H: the 2×1 dimer rows break x/y symmetry strongly, maximising E-score.
- Overall scores are close (0.78 vs. 0.75) — the exact Hamiltonian in §14 resolves which surface actually perturbs an NV more.

12 Exact Spin-Strain Hamiltonian

The screening model in §11 is superseded, where available, by the full symmetry-allowed ground-state Hamiltonian of Udvarhelyi, Shkolnikov, Gali, Burkard & Pályi, Phys. Rev. B **98**, 075201 (2018), evaluated in the $\{|+1\rangle, |0\rangle, |-1\rangle\}$ basis:

$$\mathcal{H}_0/h = [h_{41}(\varepsilon_{xx} + \varepsilon_{yy}) + h_{43}\varepsilon_{zz}] S_z^2 \quad (17)$$

$$\begin{aligned} \mathcal{H}_1/h = & \frac{1}{2} [h_{26}\varepsilon_{xz} - \frac{1}{2}h_{25}(\varepsilon_{xx} - \varepsilon_{yy})] \{S_x, S_z\} \\ & + \frac{1}{2} [h_{26}\varepsilon_{yz} + h_{25}\varepsilon_{xy}] \{S_y, S_z\} \end{aligned} \quad (18)$$

$$\begin{aligned} \mathcal{H}_2/h = & \frac{1}{2} [h_{16}\varepsilon_{xz} - \frac{1}{2}h_{15}(\varepsilon_{xx} - \varepsilon_{yy})] (S_y^2 - S_x^2) \\ & + \frac{1}{2} [h_{16}\varepsilon_{yz} + h_{15}\varepsilon_{xy}] \{S_x, S_y\} \end{aligned} \quad (19)$$

Coupling	MHz/strain	Coupling	MHz/strain
h_{41}	−6420	h_{25}	−2600
h_{43}	+2300	h_{26}	−2830
h_{15}	+5700	h_{16}	+19660

(Udvarhelyi et al., Table I, DFT/PBE values; a Barson-et-al.-scaled set is kept as a sensitivity bracket.) $\mathcal{H}_0$ is diagonal (A_1 strain: pure axial shift). $\mathcal{H}_1$ mixes $m_s=0$ with $m_s=\pm 1$ — symmetry-forbidden in bulk, allowed once the surface breaks it, and contributing only at second order, $\sim (h\varepsilon)^2/D_0$. $\mathcal{H}_2$ is the direct $m_s=+1 \leftrightarrow -1$ transverse splitting. Because $\mathcal{H}_1$ destroys the block-diagonal structure, D and E are not read off the diagonal: the full 3×3 matrix $D_0 S_z^2 + \mathcal{H}_0 + \mathcal{H}_1 + \mathcal{H}_2$ is diagonalised exactly, and the $m_s=0$ -like eigenstate is identified by maximum overlap $|\langle 0|v_i\rangle|^2$ rather than by eigenvalue order.

13 From Slab Strain to NV Strain

Two steps connect the slab’s measured in-plane strain to the strain seen by an NV.

1. Traction-free elasticity. The strain fit (§9) gives only $\varepsilon_{xx}, \varepsilon_{yy}$. A free surface requires $\sigma_{i3} = 0$; with the cubic stiffness C_{ijkl} ($C_{11}=1076$, $C_{12}=125$, $C_{44}=576$ GPa) rotated into the slab frame $C' = R \otimes R \otimes R \otimes R : C$, this is a 3×3 linear solve for $(\varepsilon_{xz}, \varepsilon_{yz}, \varepsilon_{zz})$:

$$\sigma_{i3} = C'_{i3kl} \varepsilon_{kl} = 0, \quad i = 1, 2, 3. \quad (20)$$

Using the full anisotropic tensor (not an isotropic Poisson shortcut) matters on (110)/(111), where shear components are symmetry-allowed.

2. Orientation projection. Diamond has four inequivalent $\langle 111 \rangle$ NV axes. Each sees the same crystal strain differently once rotated into its own local frame F (generated by the C_2 rotations that are true diamond-lattice symmetries):

$$\varepsilon^{(\text{NV})} = F \varepsilon^{(\text{cubic})} F^\top. \quad (21)$$

The four resulting Hamiltonians (§12) are diagonalised independently, giving a predicted zero-field stick spectrum per face rather than a single number.

14 Exact NV Predictions for H-Terminated Slabs

Using the biaxial equilibrium strain of §9 (the slab’s actual relaxed state) through §12–§13:

Surface	NV group	ΔD (MHz)	E (MHz)	f_+ (MHz)	f_- (MHz)
C(100)–H	all 4 (degenerate)	+30.9	22.6	2923.4	2878.3
C(110)–H	pair A (2 axes)	−62.3	36.4	2844.1	2771.2
C(110)–H	pair B (2 axes)	−20.9	10.2	2859.3	2839.0
C(111)–H	on-axis [111]	−11.0	0.0	2859.0	2859.0
C(111)–H	inclined (3 axes)	−3.9	4.0	2870.2	2862.1

Interpretation.

- C(100): all four axes are equivalent under equal-biaxial (100) strain, yet none lies along the surface normal — so every orientation still picks up nonzero E (isotropic in-plane, but anisotropic in/out-of-plane, strain looks anisotropic once rotated into a tilted NV frame).
- C(110): genuinely two inequivalent pairs — an experimentally distinguishable 4-line stick spectrum, not 2.
- C(111): on-axis NV has $E \equiv 0$ exactly (C_{3v} -protected); the three inclined NVs are degenerate with common, nonzero E — the model’s built-in falsifiable signature.
- Ranking by predicted line spread revises §11: $C(110) \gtrsim C(100) \gg C(111)$, whereas the cruder stress-based score (§11) had ranked C(100) marginally ahead of C(110). The exact orientation- and elasticity-resolved treatment is the one to trust.
- Magnitudes (tens of MHz) are broadly consistent with the screening model’s illustrative GHz-per-strain estimates (§11), which is expected since that model ignores orientation projection and elastic anisotropy.

Status. These are the H-terminated absolute predictions at 6 layers. The campaign’s actual target is the *difference* against matched bare/O-ether slabs at the same thickness (inputs prepared, not yet run) — the experiment compares surface states, not H vs. vacuum.

15 Convergence Report

K-Point Convergence Table

k -mesh	Energy	τ (N/m)	ΔD (MHz)
$5 \times 5 \times 1$	...	2.01	30.5
$7 \times 7 \times 1$	...	2.07	30.9
$9 \times 9 \times 1$	...	2.08	31.0
$11 \times 11 \times 1$	...	2.08	31.0

Vacuum Convergence Table

Vacuum ($\text{\AA}$)	σ (kbar)	τ (N/m)
8	35	2.04
10	28	2.05
12	23	2.05
16	17	2.06

Thickness Convergence Table

Layers	τ (N/m)	ΔD (MHz)
6	2.05	31
8	1.82	28
10	1.74	27
12	1.73	27

Plane-Wave Cutoff Convergence Table

Cutoff (Ry)	τ (N/m)
60	2.03
70	2.05
80	2.05

Fixed-Cell Convergence Table

Method	τ (N/m)
Fixed cell	2.05
Variable cell	2.09

Strain-Fit Convergence Table

Quantity	Value
σ slope	...
τ slope	scaled
σ zero-stress strain	0.0041
τ zero-stress strain	0.0041

Elastic-Tensor Convergence Table

Source	Bulk modulus (GPa)
Hydrostatic	445
Tensor	443